

Guideline Title: Bradycardia

Guideline Number: GUID-3203-C004

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
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Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Improve cardiac output in patients with impaired sinus node function or atrial/AV nodal conduction causing a disturbance in the heart's normal rhythm.
- II. **DEFINITIONS:**
 - a. Sinus bradycardia (HR <60/min, regular rhythm).
 - b. First degree heart block (prolonged PR interval > 0.20 sec).
 - c. Second degree heart block type 1 [Wenckebach] (progressive lengthening of the PR interval followed by a dropped QRS complex).
 - d. Second degree heart block type 2 (PR interval constant and set with no lengthening, some P waves will not be followed by a QRS complex).
 - e. Third degree heart block (both the atrial and ventricular rhythms are regular but independent of each other, disassociated).
- III. **DEPARTMENT GUIDELINE:**
 - a. Assess and manage airway, breathing, and circulation.
 - b. Apply cardiac monitor, confirm brady-dysrhythmia.
 - c. Obtain 12 lead ECG if available and time permits. Identify type of AV block if present.
 - d. Provide supplemental oxygen to maintain saturation > 93%.
 - e. Establish IV/IO access.
 - f. If patient is not symptomatic with stable vital signs, monitor closely.
 - g. If patient is symptomatic, differentiate between signs and symptoms that are caused by slow rate versus those that are unrelated.
 - i. Symptoms include chest discomfort or pain, shortness of breath, decreased level of consciousness, weakness, fatigue, light-headedness, dizziness, and presyncope or syncope.
 - ii. Signs include hypotension, orthostasis, diaphoresis, pulmonary congestion on physical exam or chest x-ray, congestive heart failure

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- iii. Consider **Atropine 1 mg IV/IO**, may repeat every 3-5 minutes (max. dose 0.04 mg/kg up to 3 mg). Atropine not likely to be effective in second degree type II and third-degree AV blocks.
 - h. If Atropine is ineffective, prepare patient for *Transcutaneous External Pacing per GUID3203-PR022*. Start pacing with the following settings: MA 50, HR 60. Increase MA until capture is obtained.
 - i. Considering sedation and/or analgesia per *GUID3203-G005 Analgesia and Sedation Management*.
 - i. Alternative treatments for rate-accelerating effects include
 - i. **Norepinephrine IV 0.05-1 mcg/kg/min** titrate q5 min to effect
 - ii. **Epinephrine infusion 0.025-0.5 mcg/kg/min IV** and titrate q5 min to effect
 - iii. **Dopamine infusion 5-20 mcg/kg/min IV** and titrate q5 min to effect.
 - j. If hypotension persists, consider initiating vasopressors per *GUID3203-M011 Hypotension*.
 - k. Consider alternative drugs for beta-blocker or calcium channel blocker overdoses. See *GUID3203-M019 Toxic Ingestion/Exposure*.
 - l. Consider expert consultation from sending or receiving physician.
 - m. Consider the latest recommendations from the American Heart Association.
- IV. **DOCUMENTATION:**
 - a. EMS Charts
- V. **REFERENCES:**
 - a. American Heart Association, 2015.